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Edge Processing - A Paradigm for Instantaneous Value Realization

Introduction

Industrial companies are driving new levels of performance and productivity gains, in the form of reduced unplanned downtime, higher production efficiency etc. leveraging cloud computing and other technology innovations.

A key element of industrial transformation is the speed of data and analysis. According to a [study from IDC](#), 45% of all data created by IoT devices will be stored, processed, analyzed, and acted upon close to, or at the edge of, a network by 2019. As more IoT devices get added and the need for handling time-critical use cases increases, a new paradigm is required to aggregate and process data, draw insights from, and initiate actions close to assets producing the data.

Edge Processing will become critical for handling the data deluge, reducing time-to-value and realizing value instantaneously.

This paper talks about the cloud-based approach for data processing, its challenges, and how Edge Processing addresses those needs. It concludes with how Edge and Cloud can operate together for realizing business outcomes.



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Business Imperatives, Objectives and KPIs to Measure Objectives

Broadly speaking, manufacturers have the following business imperatives and objectives:

Imperatives	OPERATING THE BUSINESS	GROWING THE BUSINESS
Objectives	Enhance Productivity Machines Processes People Reduce Risk Real-time response	Grow Revenue Find new revenue streams

Figure 1: Imperatives and Objectives of Manufacturing Businesses

Towards the objective of **enhancing Productivity** - Overall Equipment Effectiveness (OEE) is the global standard for measuring manufacturing productivity. By combining the factors of machine Availability, Performance (production rate) and production Quality, this metric identifies the percentage of manufacturing time that is truly productive. This helps organizations to gain full visibility and traceability throughout the processes, track product and production specifications, control variability in product quality, and optimize time and costs.

OEE Factor	GoalsD	Data Required	
Availability	- Increase the uptime of machines - Reduce changeover time	Primary - Equipment failure/repair - Equipment downtime/maintenance - Material shortage	Contextualized - Production Schedules - Stoppage/changeover plans - Procurement plan
Performance	- Increase the performance in available time - Reduce idling	Primary - Production Cycle times - Production Rate - Planned & Actual	Contextualized - Machine health & wear - Operating time of equipment - Material feed plans
Quality	Reduce process defects	Primary - Process defects - Total yield	Contextualized - Equipment failure & maintenance - Process updates/adjustments

Figure 2: Data (direct and contextual) Required for Measuring OEE Factors



Acquiring Data

Data on the factory floor can be acquired from sensors that are mounted on devices, controllers that are connected to devices and sensors, data historians and any local data sources.

Example: An auto plant with the objective of reducing component defects, may use sensors to **measure** 50,000 data points for each part produced. Other machines capture x-ray and heat treatment data, while separate databases track supplier data and quality data.

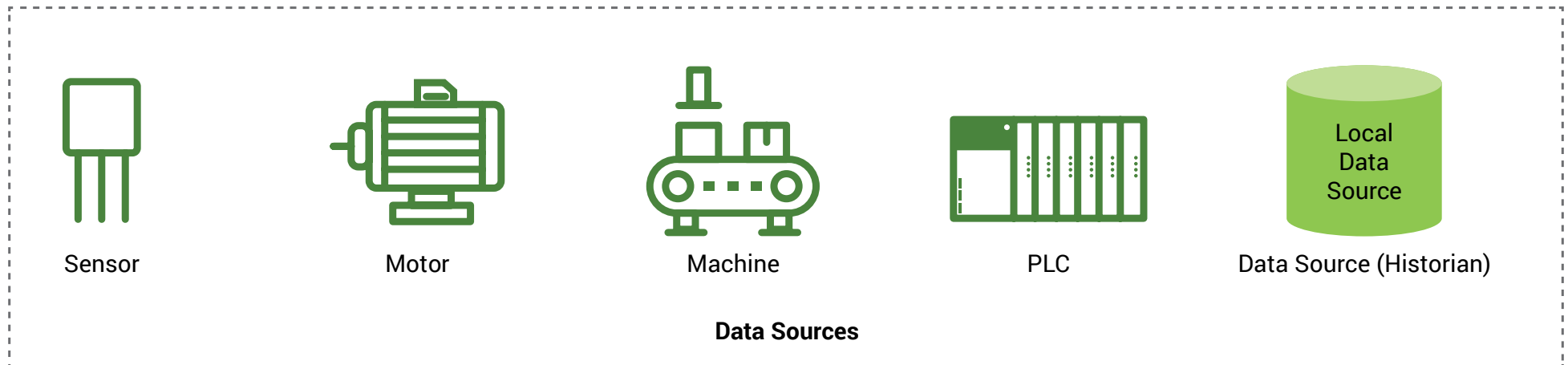


Figure 3: Sources for Industrial Process and Machine data



Processing Data

One of the approaches for processing data acquired from sensors/controllers/historian etc. is by ingesting the data to a cloud-based centralized IoT platform that can process data in real-time. The cloud-based IoT platform aggregates data from disparate data sources, applies business rules on the live feed of data, and triggers actions based on the outcome. Actions include notification to user downstream, command back to the device upstream, etc.

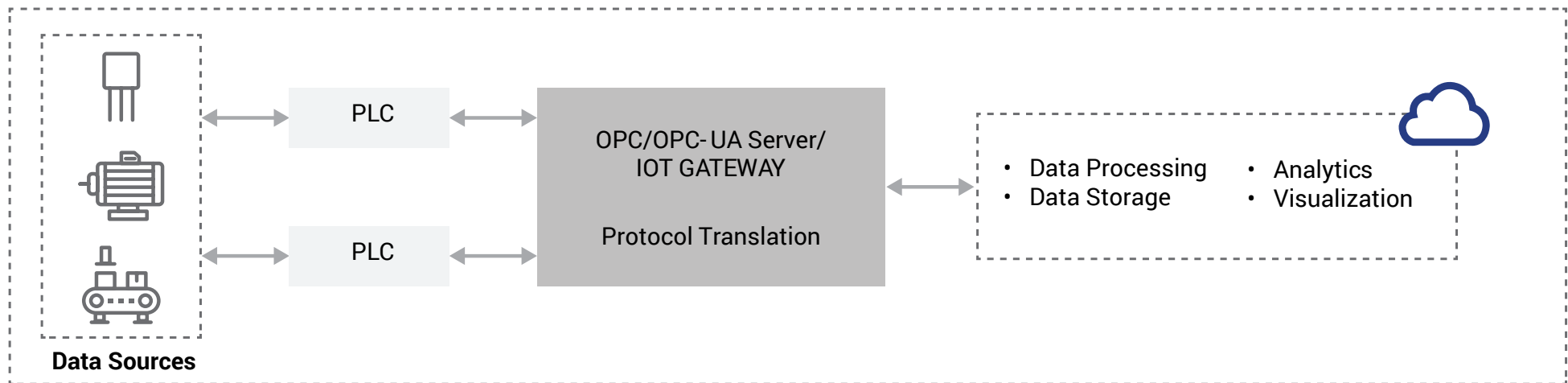


Figure 4: A Centralized Data Processing system



Challenges with the approach

Cloud-based data processing leverages a centralized networked storage and computing capability of systems to deliver the necessary outcome. A critical success factor for this approach is the ubiquitous availability of network bandwidth and low latency. However, manufacturing plants and enterprises face challenges like limited network connectivity, high latency, rising storage and processing costs, and potential security breach.

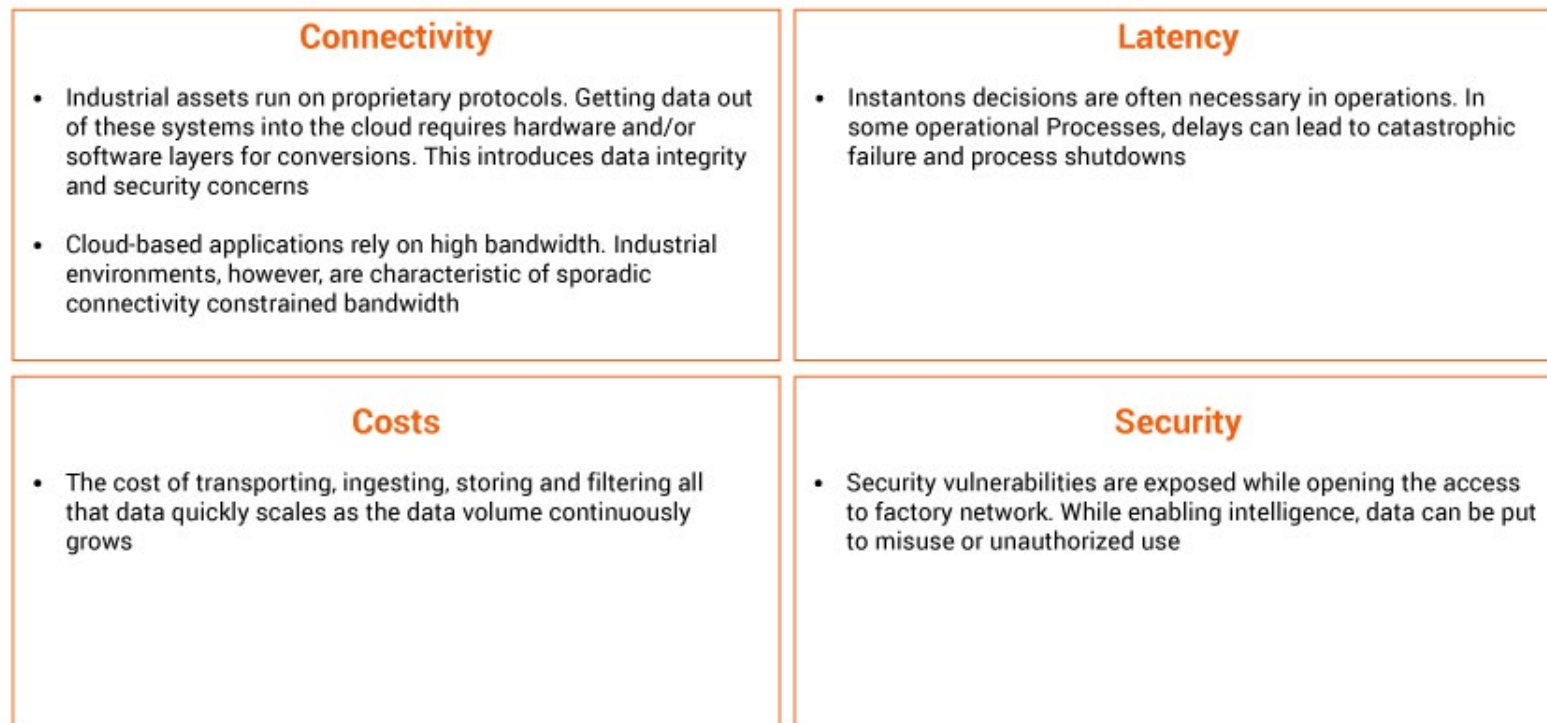


Figure 5: Challenges in a centralized data processing system



Here are some scenarios depicting the challenges arising in a centralized data processing set-up.

Example 1 - Protecting equipment from damage by overheating

A Thermocouple measures temperature on a pump/motor. When it is determined that the temperature has exceeded the defined threshold, the pump should be shut down in milliseconds without any decision latency. The time value of the temperature information decays rapidly as delayed response can result in damage.

Example 2 - Monitoring the Performance of Production Lines

The performance of production lines is expressed through indicators like OEE. Real-time analysis of multiple data points is required to provide OEE trends and alerts to operational personnel. The time value of information is high as response delays can cause significant losses.

Example 3 - Reducing Safety Risks

According to an [estimate](#), an offshore oil platform generates between 1 TB and 2 TB of time-sensitive data related to production and drilling safety per day. With satellite communication, the data speeds range from 64 kbps to 2 Mbps. This results in 12 days to transmit one day's worth of data back to a central site for processing and could have significant operational and safety implications.

Time-Value graph

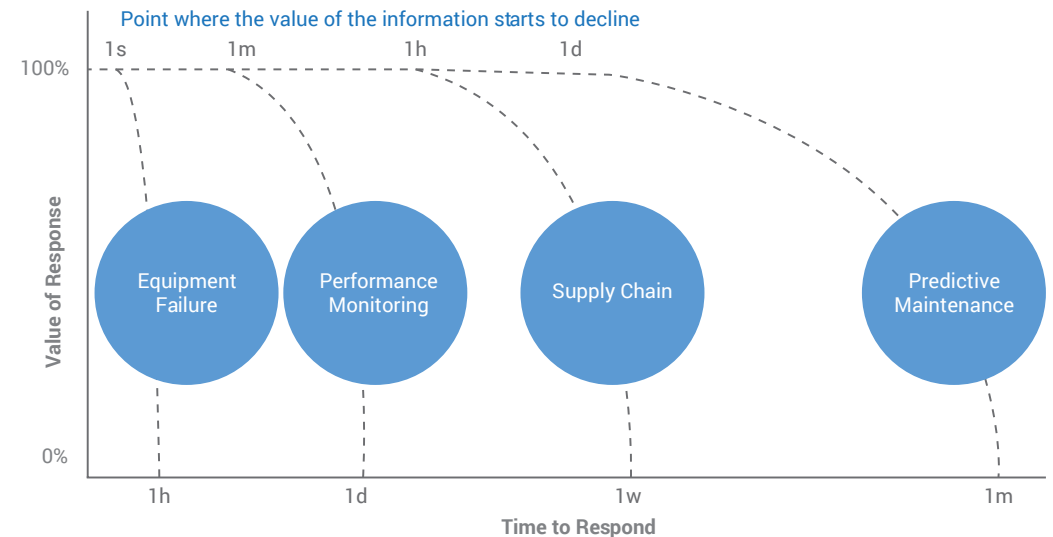


Figure 6: Rate of Information Decay depending on Time to Response and Value of Response

Image Source: [Introduction to Edge Computing in IIoT by the Industrial Internet Consortium](#)



Edge Processing - A New Paradigm for Data Processing

A framework for measuring and monitoring productivity, reducing cascading failures, and responding to events in real-time calls for a decentralized model with distributed storage, processing, analysis, decision making, and control. In this new paradigm, data is processed right where it is produced and sent to the cloud selectively.

Depending on where the data is processed, Edge Processing can be done at the controller or at the gateway.

Edge processing at the Controller

- The intelligence, processing power, and communication capabilities are directly embedded into devices like programmable automation controllers (PACs)
- Physical assets (pumps/motors/generators etc.) are physically wired into a control system where the PAC automates

them by executing an onboard control logic

- PACs can be programmed to collect, analyze, and process data from the physical assets they are connected to
- Intelligence is pushed to the network edge, where physical assets or things are first connected and where IoT data originates

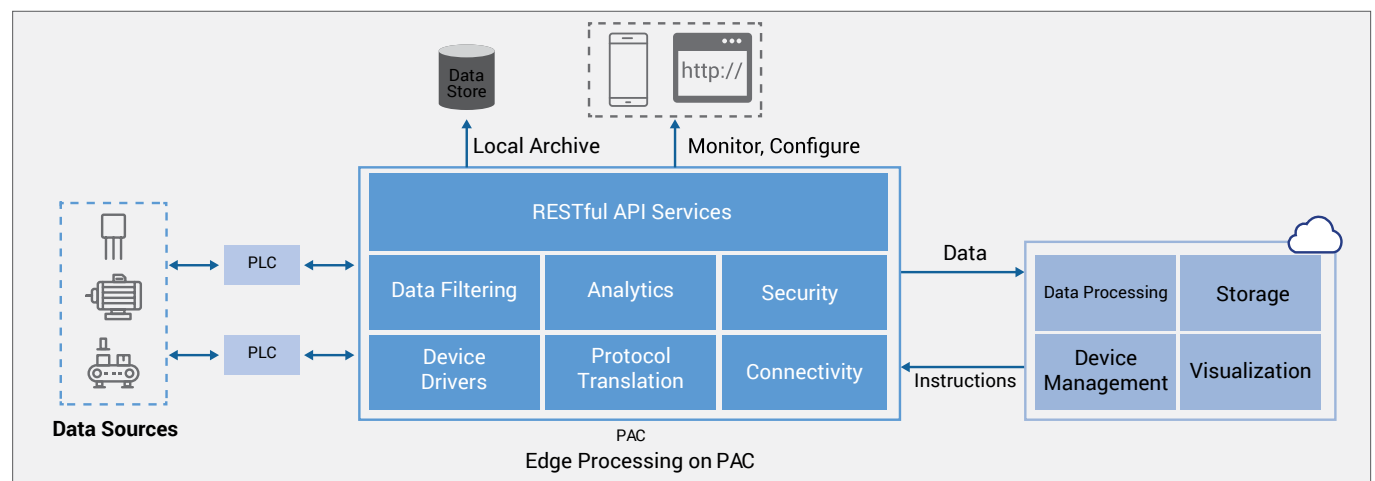


Figure 7: A Functional overview of Industrial PC based Edge processing



Edge processing at the Gateway

- The intelligence, processing power and communication capabilities are pushed to the local area network in an IoT gateway
- The data from the control system is sent to an OPC server, which converts the data into a protocol such as MQTT
- The translated data is sent to an IoT gateway on the LAN, which collects the data and performs higher-level processing and analysis. The gateway filters, analyses, processes, and stores the data for transmission to the cloud

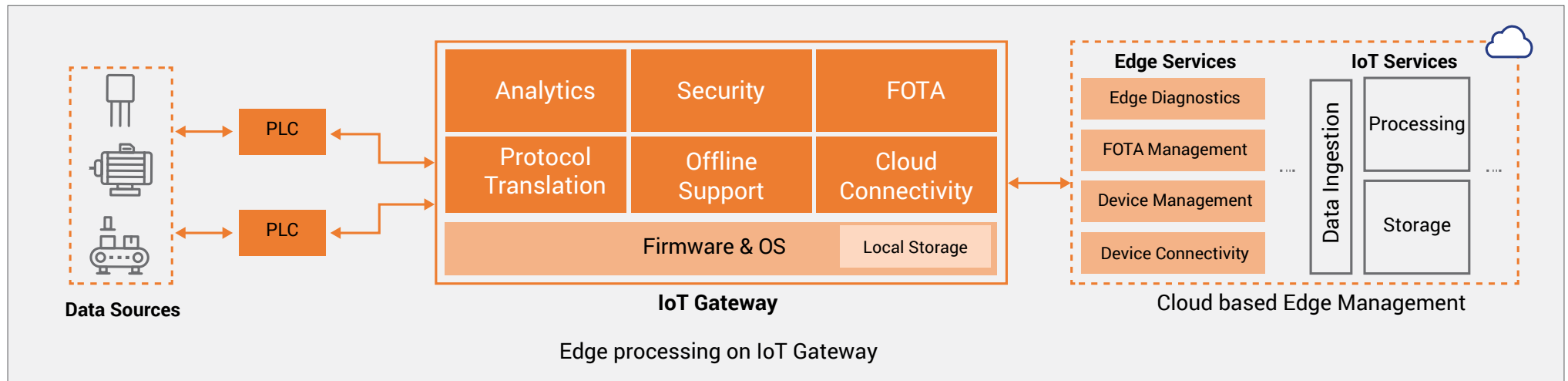


Figure 8: A Functional overview of IoT Gateway based Edge processing



In addition to enabling device interoperability, reducing latency, enhancing data security and obviating the need for high network bandwidth availability, each of the models is uniquely placed to address the challenges associated with centralized cloud-based processing:

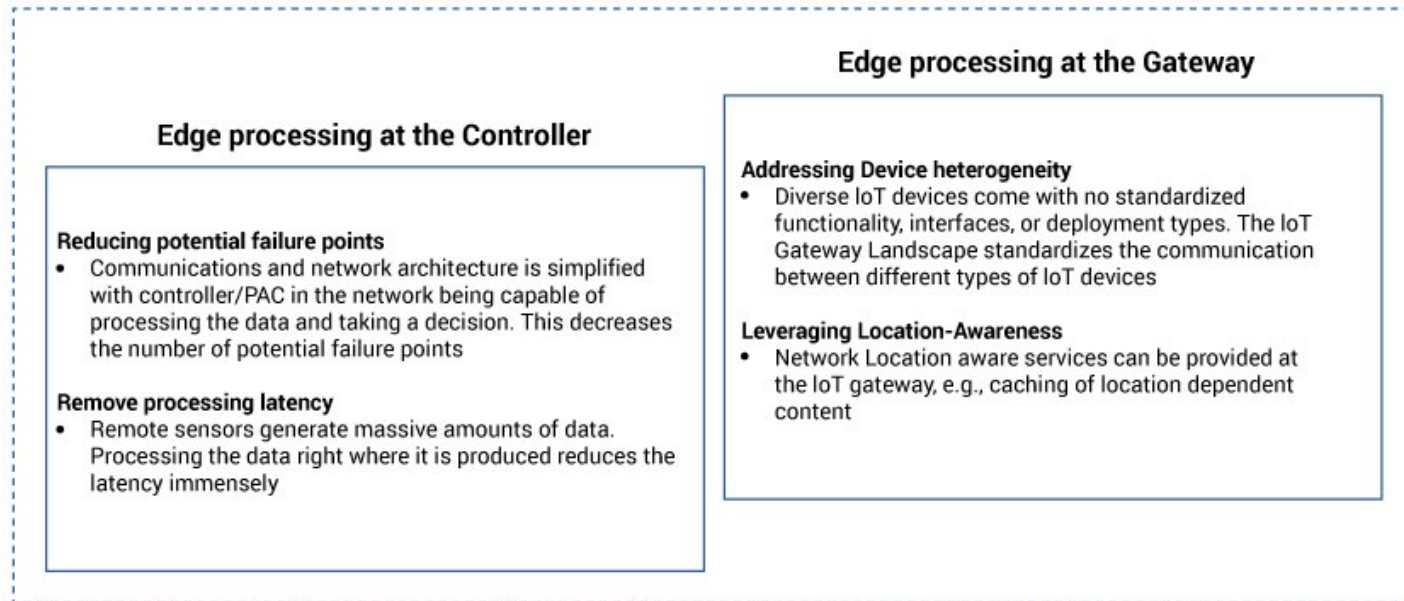


Figure 9: Characteristics of different types of Edge processing

Based on the requirements of the problem at hand, the Edge can move along the continuum of capabilities for an IIoT solution. The potential deployment scenarios are:

- Edge processing embedded within the equipment, Gateway or Industrial PC
- On-premise data center at the Plant level
- IoT Cloud at Enterprise level



Is Edge Processing the panacea for all industrial scenarios?

Complex statistical analyses, references to historical data, contextualization with process and operations, correlation across data variables and advanced visualization require large storage and processing capacity and are better off done on a centralized, scalable cloud-based IoT platform.

Sample scenarios that require cloud-based storage and processing include:

- Predictive analytics to determine whether an engine is about to fail based on sensor data gathered over the past month
- Root-cause analysis to determine why an engine has overheated rather than just indicating it's overheating

These strategic processes are better placed in the cloud that can store and process large amounts of data



Driving Business Value by Combining Capabilities of Edge and Cloud

An integrated approach for data processing leverages the capabilities of Edge for handling time-critical decisions and the Cloud for long-term storage, statistical performance modeling and data visualizations. Executing this approach requires a set of integrated, standards-based software capabilities in the form of a cloud-based IoT Platform which should:

- Be a set of loosely coupled services with storage and computing capabilities extended from the cloud to devices, and the edge
- Delegate to Edge the aspects of interoperability, responding to events in real-time, supporting offline interactions, facilitating machine-to-machine communication, securing the data transfer

from the factory floor to the cloud

- Maintain a digital twin for each of the devices and gateways in the cloud to enable device management, remote monitoring and control of operations
- Include the aspects of device management, data management,

enterprise integrations, and advanced analytics in cloud-based processing

- Complement the Edge to leverage data optimally and foster data-driven real-time decision making

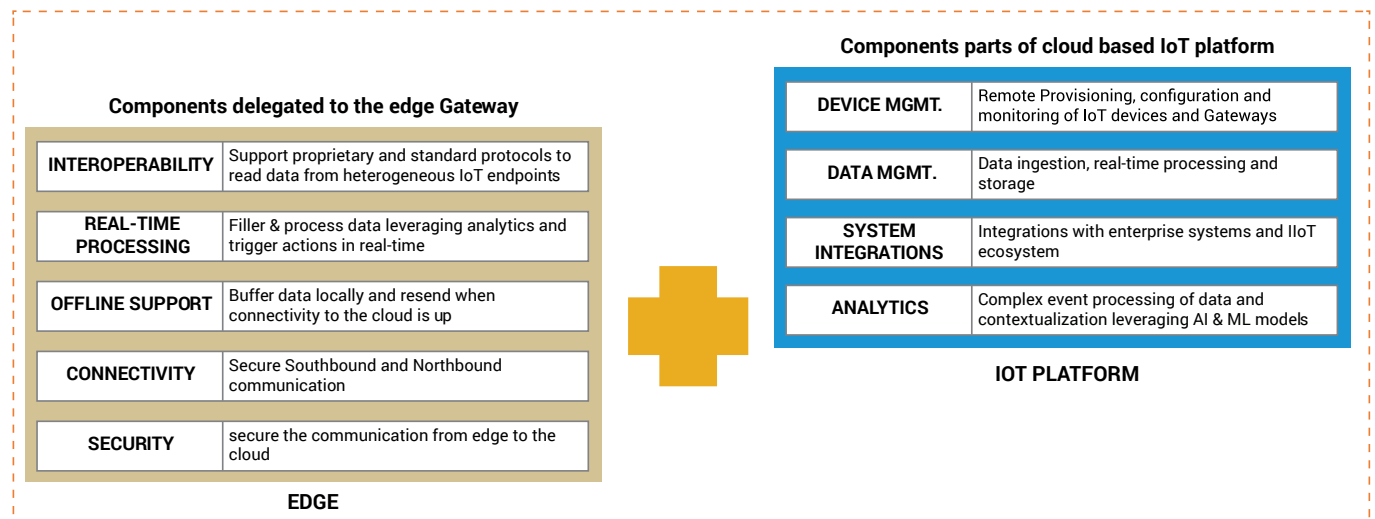


Figure 10: Industrial IoT capabilities distributed across the Edge and Cloud



A framework for Distributed Data Processing towards the objective of Enhancing Productivity

To recall, OEE is a standard KPI to measure manufacturing productivity. Here is an illustration of the goals for each of the OEE factors, and how the processing can be distributed to accomplish these goals.

OEE Factor	Goals	Processing	
Availability	• Detect machine failure • Reduce unplanned Downtime • Minimize changeover time • Alert material shortage	Edge	Cloud-based IoT platform
		• Anomaly detection • Dynamic process configurations • Alerts on machine behavior • Alerts on material shortage	• Machine Learning based performance modeling • Correlations with contextual data • Integrations with supplier and procurement systems
Performance	• Detect machine wear • Alert material quality • Standardize process changes	Edge	Cloud-based IoT platform
		• Performance monitoring • Real-time notifications	• Computation of Remaining useful Life of machines • Analytics to predict performance based on material feed quality
Quality	• Reduce rework	Edge	Cloud-based IoT platform
		• In situ quality inspection • Process collaboration • Real-time alerts	• Machine Learning based models for predicting quality • Storage of raw and processed data for audit trail

Figure 11: Distributed Data Processing for measuring OEE factors



Representative Architecture for Distributed Data Processing

Following is a representative architecture with processing distributed across the Edge and the Cloud

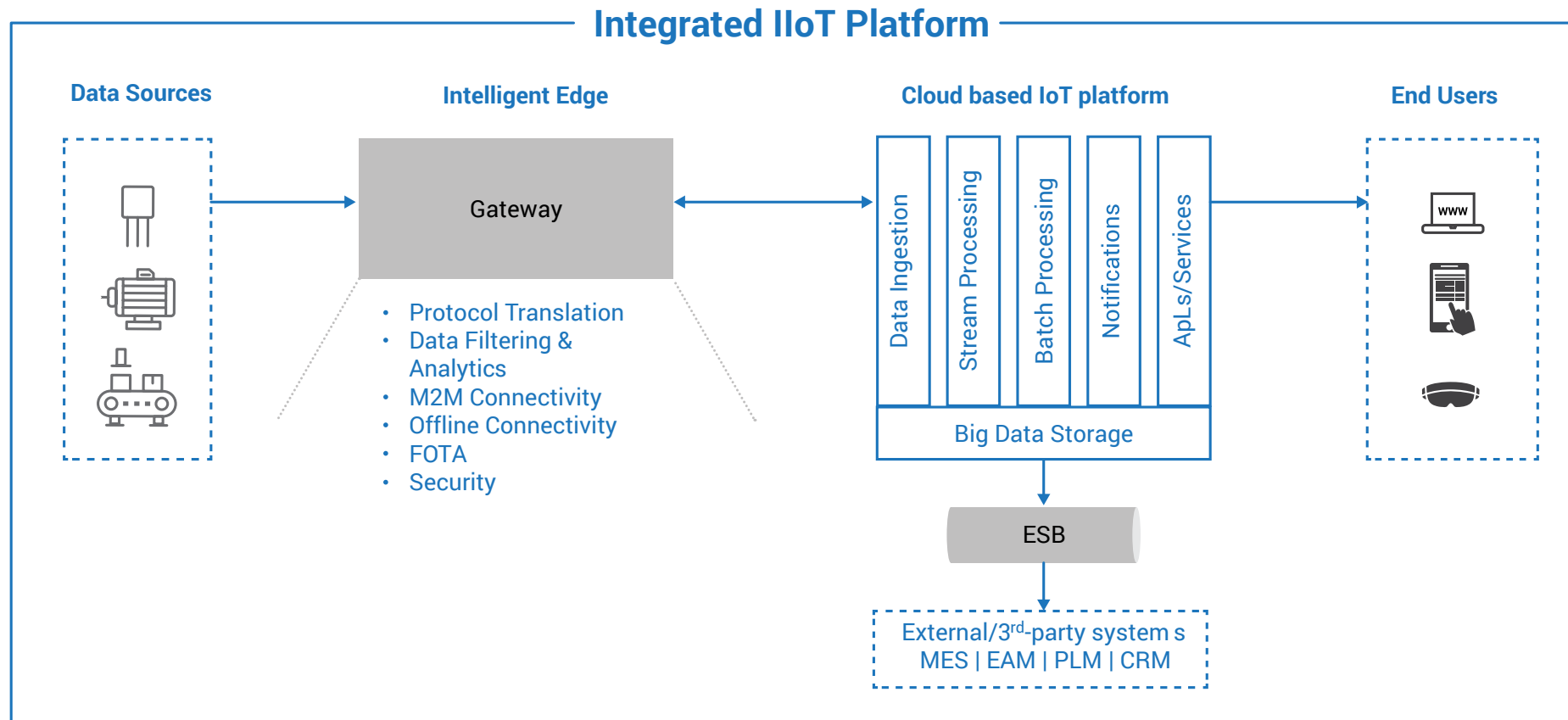


Figure 12: Overview of Industrial IoT platform complementing the Edge



Conclusion

Edge Processing accelerates awareness and response to events by eliminating a round trip to the cloud for analysis. It avoids the need for costly bandwidth additions by offloading gigabytes of network traffic from the core network. It also protects sensitive IoT data by analyzing it within company walls. Ultimately, organizations that adopt Edge Processing gain deeper and faster insights, leading to increased business agility, higher service levels, and improved safety

The IIoT platform, along with the IoT Edge, and through enterprise IT and OT integration illuminates operational visibility, enhances data availability, access for production and business stakeholders and partners, thereby facilitating data-driven decision making. This drives manufacturing and industrial industries to become digital businesses.

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About the Author

Asgar Ali has over 10 years of experience in designing, developing, and delivering Enterprise solutions for the Oil & Gas industry. At Sasken, he is responsible for building the value proposition and marketing the Digital Services for Industrial and Transportation segments.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Smart Devices & Wearables, Enterprise Grade Devices, SatCom, and Transportation industries. For over 29 years and with multiple patents, Sasken has transformed the businesses of over a 100 Fortune 500 companies, powering over a billion devices through its services and IP.

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